TypeCat

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Frontend

Click Events → API
Login, Register, Submit Level, Load Dashboard, Load Leaderboard

[(ngModel)] Forms
Login: username, password · Register: username, email, password, avatar

Named Routes
/login · /register · /dashboard · /leaderboard · /play/:id

@for / @if
Level list with @for · Auth state & errors with @if

JWT Auth
HTTP interceptor attaches Bearer token · authGuard protects routes · logout clears storage

Angular Services
AuthService · GameService — all HTTP via HttpClient, centralized

Error Handling
API errors caught and displayed as user-facing messages on all failed requests

CSS Styling
Global SCSS with dark theme, gradients, animations · Nginx production serve

BACKEND

≥ 4 Django Models
User · ProfileImage · Level · Submit · Balance · Transaction

ForeignKey Relations
User → ProfileImage · Submit → Level · Transaction → Balance

Serializers
LoginSerializer, SubmitSerializer (Serializer) · UserModelSerializer, LevelModelSerializer (ModelSerializer)

FBV + CBV Views
@api_view: token verify, refresh · APIView: Login, Register, Balance, Submit, Leaderboard

JWT Auth
POST /auth/login → access (5 min) + refresh (7 day) tokens · Traefik ForwardAuth middleware

Full CRUD — Level
GET /level (list) · GET /level/:id (read) · POST /level (submit) · Admin: Update, Delete

Linked to Auth User
Submit stores user_id from X-User-Id header injected by Traefik after auth verification

CORS Configured
django-cors-headers allows http://localhost:4200 — Angular dev server fully connected

System Design

MICROSERVICES

auth_service
Identity & JWT · PostgreSQL
level_service
Levels & Scoring · PostgreSQL
balance_service
Wallet & Credits · PostgreSQL
leaderboard_service
Daily Rankings · Redis ZSET

KAFKA EVENT FLOW

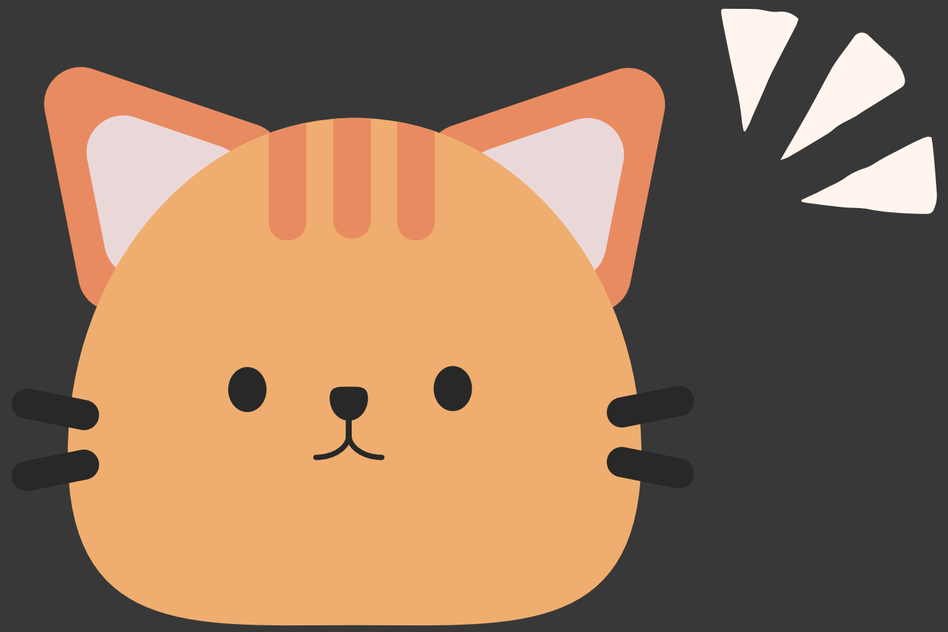
auth_service → user.registered → balance_service
Debezium CDC → dbz.public.users → balance_service
level_service → submit.rewarded → balance_service
level_service → submit.rewarded → leaderboard_service

SECURITY — TRAEFIK FORWARDAUTH

Browser → :8000 Traefik (only public entry)
Traefik → GET /auth/verify (every protected request)
Auth returns X-User-Id + X-Username headers
Services read headers — no JWT parsing needed

INFRASTRUCTURE

PostgreSQL ×3
Redis 7
Kafka
Debezium CDC
MinIO (S3)
Prometheus
Grafana
Traefik
Docker Compose



Thank You!